322 Midterm Summative Activity: Classification Models

Objectives:

1. Develop classification models to predict a target variable.
2. Evaluate the classification models based on the different performance metrics.

This activity is to be accomplished by group.

Tasks (Use Python)**:)**

1. Choose an appropriate dataset that you can use to analyze the patterns in which a target variable can be predicted. Ensure that the dataset is at least 1000 rows after data preprocessing. Use a rule-based classification (ex: Decision trees)

2. Using the same dataset, develop classification models using the following algorithms:
   a. Naive Bayes
   b. Support Vector Machine
   c. your own choice of classification algorithm

   For each algorithm
   - Split the cleaned dataset into 90-10 proportion:
     - 90% - training & testing data
     - 10% - unseen data

     For the training and testing dataset, split into 80-20 proportion
     - 80% - training
     - 20% - testing
   - Apply ten-fold cross-validation and evaluate the algorithms using:
     a. Confusion matrix
     b. Accuracy
     c. Precision
     d. Recall
     e. F1-score
     f. Specificity
     g. ROC-AUC

   - Analyze the results of each classification algorithm. Analysis must also include the comparison of the performance of the classification algorithms and which algorithm/s are most appropriate for the chosen dataset
   - Validate your models using the unseen data. Discuss the results.

Deliverables:

1. Dataset in csv format (Filename: Team name-MidtermSummativeAct.csv)
   Note: present the raw dataset up to the cleaned dataset in tabs (if applicable - if the group collected a dirty dataset)

2. Prepare a document with the contents described below:
   ○ example filename: *Team name-MidtermSummativeAct.pdf*

a. **Cover page.** Provide the title of your activity, class code, group name and alphabetical list of members

b. **Objectives.** State the objectives of your analysis in the context of your dataset

c. **Methods**. Provide the logical and systematic presentation of the methods used to achieve the objectives of this activity. The methods include the set of steps of activities taken, starting from data exploration up to the completion of the activity. Discuss the tools and technologies utilized to implement the methods.

. For the presentation of the Python codes, present all codes , output (for the output, use white background) and remarks(use of the code, not the interpretation) in a tabular format (similar to previous exercises/activities). Provide delimited comments to the Python codes.

d. **Results and Discussion of results** - this section must present the results of the methods discussed in the previous section.

- present the discussion of the main results of the predictive analysis.

- data preprocessing
- the patterns in which a target variable can be predicted for the rule-based classifier
- classification models using the following algorithms:
    a. Naive Bayes
    b. Support Vector Machine
    c. your own choice of classification algorithm

- present the performance metrics of each classification algorithm in tabular form
- provide the analysis of the classification evaluation metrics
- Which algorithm yielded the best performance?

f. **References**

- include AI use and percentage
- other references

h. **Appendix**

- links to your dataset and Python code
- peer accomplishment report

Peer Accomplishment Report

○ example filename: *Team name-MidtermSummativeAct-Peer.pdf*

Name	Recent photo	Accomplishment	% Contribution

Provide detailed descriptions of the member's accomplishment. The % contribution must have a total of 100%.

4. Submit your document in the designated Google classroom assignment bin.
5. Submit the hardcopy of your output in our lab class.

Q&A:

This will be an open forum where the instructor will ask questions to the team to validate the output and the team members' knowledge of the activity.

Criteria

Rating Scale:
5 - Excellent, 4 - Very Good, 3 - Adequate, 2 - Needs Improvement, 1 - Non-compliant, 0 - No output

Criteria	Weight
Methods (Data mining framework)	x 2
Results & Discussion of results	
Data preprocessing	x 1
Rule-based classifier • Code & Output • Analysis	x 3

Naive Bayes
• Code & Output
• Performance metrics
  • Confusion matrix
  • Accuracy
  • Precision
  • Recall
  • F1-score
  • Specificity
  • ROC-AUC
• Validation
• Analysis
	Code & output (x 2)
	Performance Metrics (x 2)
	Validation (x 1)
	Analysis (x 2)

Support Vector Machine
• Code & Output
• Performance metrics
  • Confusion matrix
  • Accuracy
	Code & output (x 2)
	Performance Metrics (x 2)
	Validation (x 1)
	Analysis (x 2)

● Precision ● Recall ● F1-score ● Specificity ● ROC-AUC ● Validation ● Analysis	
Chosen algorithm ● Code & Output ● Performance metrics ● Confusion matrix ● Accuracy ● Precision ● Recall ● ROC-AUC ● Validation ● Analysis	Code & output (x 2) Performance Metrics (x 2) Validation (x 1) Analysis (x 2)
Comparison of algorithms	x 2
Q&A	X2
Total Score	
	over 155

This laboratory activity shall be evaluated and graded based on your compliance with the activity guidelines.